



Conductivity Calibration Certificate

RBRlegato³ C.T.D, Teledyne Webb Slocum, dry bay (1000dbar) s/n: 205806

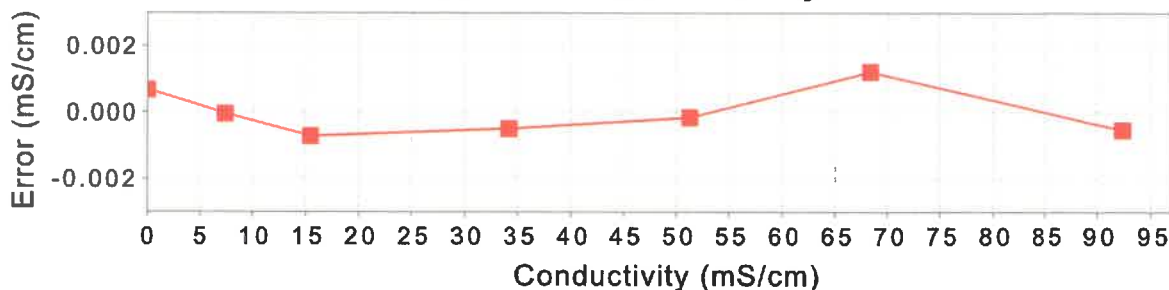
References: Autosal8400B#66289, MS-315#15506, SSW P162, RC#002

Reference Resistance (ohm)	Reference Conductivity (mS/cm)	Voltage Ratio, V	Measured Conductivity (mS/cm)	Calibration Error (mS/cm)	Coefficients	
open	0.0000	-0.000166	0.0007	0.0007	C0:	32.019407E-3
694.042	7.3896	0.038896	7.3896	-0.0000	C1:	189.15787
331.926	15.4514	0.081512	15.4507	-0.0007	(K) C2:	1.001942
150.019	34.1871	0.180561	34.1866	-0.0005	X0:	810.437E-6
100.016	51.2792	0.270922	51.2791	-0.0001	X1:	-19.002095E-6
75.023	68.3617	0.361237	68.3629	0.0012	X2:	0.0
55.520	92.3769	0.488187	92.3763	-0.0005	X3:	0.0
					X4:	0.0
					X5:	15.012333
					X6:	10
Bath	Voltage Ratio	Temperature (ITS-90)	Salinity (PSS-78)	Conductivity (mS/cm)		
T15S35	0.2268530	15.01233	35.0122	42.9431		
T25S35	0.2747243	23.96505	35.0053	51.9999		

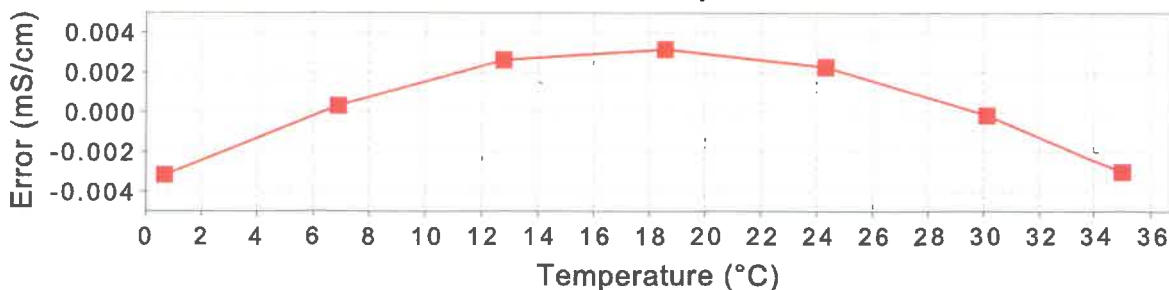
Cell Constant @T15S35 = 5.12872 1/cm

$$C_c = \frac{C_0 + C_1 * C_2 * V - X_0 * (T - X_5)}{1 + X_1 * (T - X_5) + X_2 * (P - X_6) + X_3 * (P - X_6)^2 + X_4 * (P - X_6)^3}$$

Calibration error vs. Conductivity



Calibration error vs. Temperature



Calibration Date: 2020-12-05
Issue Date: 2020-12-05
File Name: 205806_20201205_1709C.rsk

Operator: Jeff Walker
jwalker

Approver: I. Shkvetsov
ishkvorets

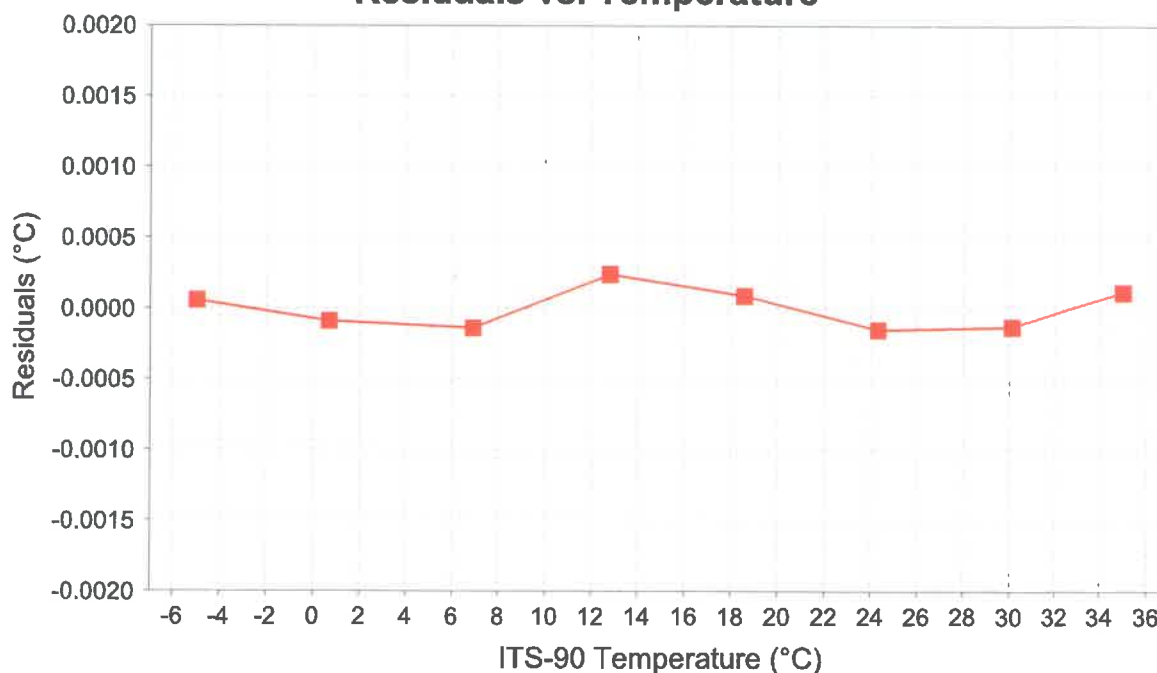
RBR

Temperature Calibration Certificate

Logger ID: RBRlegato³ Serial No: 205806 Channel No: 2

Reference Temperature, ITS-90	Voltage ratio, V	Measured Temperature, ITS-90	Calibration error	Coefficients	
-4.97276	0.715388	-4.97271	0.00006	C0:	3.494288E-3
0.67187	0.650599	0.67178	-0.00009	C1:	-252.0697E-6
6.89749	0.575099	6.89735	-0.00014	C2:	2.585521E-6
12.80477	0.502742	12.80501	0.00024	C3:	-75.74677E-9
18.57310	0.434367	18.57319	0.00009		
24.31789	0.370800	24.31774	-0.00015		
30.11202	0.312819	30.11189	-0.00013		
34.99734	0.269317	34.99746	0.00012		

Residuals vs. Temperature



Calibration Date: 2020-11-27
Issue Date: 2020-11-28
Calibration ID: 42723

Operator: 
dluong

Approver: 
kmalorny



Pressure Calibration Certificate

RBRlegato³ C.T.D, Teledyne Webb Slocum, dry bay (1000dbar) s/n: 205806

Sensor rating: 1000 dbar s/n: L162825

Nominal accuracy: 0.05%FS (0.5 dbar)

Reference instrument: Mensor CPC6050 s/n: 41000CAM

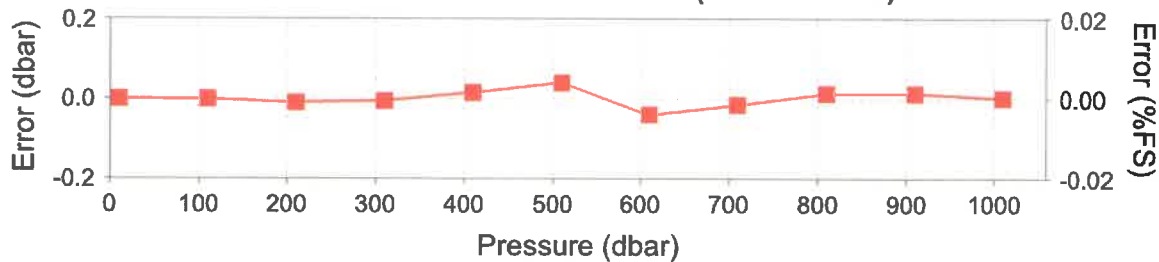
Applied pressure, P _{app} (dbar)	Voltage ratio, V	Measured pressure, P _c (dbar)	Calibration error (dbar)	Coefficients
10.026	0.016595	10.0261	0.0006	C0: -27.866503
110.000	0.060056	109.9998	-0.0002	C1: 2.2970085E3
209.998	0.103471	209.9886	-0.0094	C2: 41.573944
310.002	0.146844	309.9966	-0.0054	C3: -33.24471
409.997	0.190176	410.0125	0.0155	X0: 10.0279
510.000	0.233475	510.0401	0.0401	X1: 54.021347E-3
609.995	0.276696	609.9561	-0.0389	X2: -75.95902E-6
710.003	0.319945	709.9883	-0.0147	X3: 66.91884E-9
809.998	0.363175	810.0115	0.0135	X4: -119.96945E-6
910.001	0.406386	910.0140	0.0130	X5: 21.818092
1010.000	0.449589	1010.0015	0.0015	

$$P_c = X_0 + \frac{P_m - X_0 - X_1(T - X_5) - X_2(T - X_5)^2 - X_3(T - X_5)^3}{1 + X_4(T - X_5)}$$

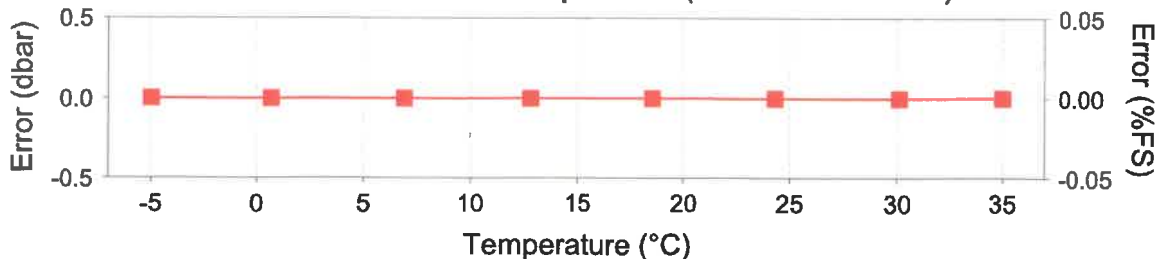
Head (mm) = 230

$$P_m = C_0 + C_1V + C_2V^2 + C_3V^3$$

Calibration error vs. Pressure (Tcal = 21.8°C)



Calibration error vs. Temperature (Patm = 10.05 dbar)



Calibration Date: 2020-12-03

Issue Date: 2020-12-03

File Name: 205806_20201203_1705P.rsk

Operator:

T. Aduetteh
takuetteh

Approver:

K. Malorny
kmalorny

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1  RBRCTD-205806, 04/29/2021, MJB
2
3  SciShell> uart j2 9600 0
4
5  bit_raise: Raising bit(0).
6  Switching to J2
7  Type ctrl-c to return to glider
8  -----
9  2000-01-01 00:00:02.000, 0.0008, 23.9488, 10.0081, -0.1243, -0.1235, 0.0115,
10 1.0000, 23.8137
11 2000-01-01 00:00:02.500, 0.0010, 23.9483, 10.0061, -0.1263, -0.1255, 0.0115,
12 1.0000, 23.8137
13 2000-01-01 00:00:03.000, 0.0012, 23.9478, 10.0116, -0.1209, -0.1201, 0.0115,
14 1.0000, 23.8731
15
16
17
18 GliderLAB I -3 >loadmission sci_on.mi
19
20 load_mission(): Opening Mission file: sci_on.mi
21 Setting SENSOR c_science_on(enum) value 3.000000
22 Setting SENSOR c_science_all_on(secs) value 0.000000
23 Setting SENSOR c_science_send_all(bool) value 1.000000
24 Setting SENSOR u_use_ctd_depth_for_flying(bool) value 0.000000
25
26 GliderLAB I -3 >124.40 70 Sent science:s:c_profile_on(sec) 0-
27 124.40 Sent science:s:c_science_send_all(bool) 1-
28 124.40 Sent science:s:m_cycle_number(nodim) 70-
29 127.21 science wrote:sci_m_present_secs_into_mission(sec) 126.710999965668
30 127.21 science wrote:sci_m_present_time(timestamp) 1619718970.241
31 128.44 71 Sent science:s:m_cycle_number(nodim) 71-
32 131.25 science wrote:sci_m_present_secs_into_mission(sec) 130.713000059128
33 131.25 science wrote:sci_m_present_time(timestamp) 1619718974.243
34 132.48 72 Sent science:s:m_cycle_number(nodim) 72-
35 134.93 science wrote:sci_rbrctd_cond_cell_temp_00(degC) 23.8731
36 134.93 science wrote:sci_rbrctd_conductivity_00(mS/cm) 0.0001
37 134.93 science wrote:sci_rbrctd_count_00(counts) 1
38 134.93 science wrote:sci_rbrctd_depth_00(m) -0.121
39 134.93 science wrote:sci_rbrctd_pressure_00(dbar) 10.0106
40 134.94 science wrote:sci_rbrctd_salinity_00(psu) 0.0116
41 134.94 science wrote:sci_rbrctd_seapressure_00(dbar) -0.1218
42 134.94 science wrote:sci_rbrctd_temperature_00(degC) 23.8336
43 134.94 science wrote:sci_rbrctd_timestamp(timestamp) 1619718977.917
44 134.94 science wrote:sci_water_cond(S/m) 0.99999998551134E-05
45 134.95 science wrote:sci_water_pressure(bar) -0.01218
46 134.95 science wrote:sci_water_temp(degC) 23.8336
47 135.21 science wrote:sci_m_present_secs_into_mission(sec) 134.714999914169
48 135.21 science wrote:sci_m_present_time(timestamp) 1619718978.245
49 135.42 science wrote:sci_rbrctd_cond_cell_temp_00(degC) 23.8731
50 135.42 science wrote:sci_rbrctd_conductivity_00(mS/cm) -0.0001
51 135.42 science wrote:sci_rbrctd_count_00(counts) 1
52 135.42 science wrote:sci_rbrctd_depth_00(m) -0.1205
53 135.42 science wrote:sci_rbrctd_pressure_00(dbar) 10.0112
54 135.43 science wrote:sci_rbrctd_salinity_00(psu) 0
55 135.43 science wrote:sci_rbrctd_seapressure_00(dbar) -0.1213
56 135.43 science wrote:sci_rbrctd_temperature_00(degC) 23.8342
57 135.43 science wrote:sci_rbrctd_timestamp(timestamp) 1619718978.416
58 135.44 science wrote:sci_water_cond(S/m) -0.99999998551134E-05
59 135.44 science wrote:sci_water_pressure(bar) -0.01213
60 135.44 science wrote:sci_water_temp(degC) 23.8342
61 135.90 science wrote:sci_rbrctd_cond_cell_temp_00(degC) 23.9028
62 135.90 science wrote:sci_rbrctd_conductivity_00(mS/cm) -0.0002
63 135.91 science wrote:sci_rbrctd_count_00(counts) 1
64 135.91 science wrote:sci_rbrctd_depth_00(m) -0.123

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PRESSURE TRANSDUCER CALIBRATION DATA

<i>Customer</i>	<i>Date</i>
TELEDYNE BENTHOS	9 JUL 21
<i>Model Number</i>	<i>Serial Number</i>
141698-2000A	129989

<i>Diaphragm Materials</i>	<i>Excitation</i>	<i>Pressure Range</i>	<i>Excitation Type</i>
TITANIUM	5 VDC	2000 PSIA	Constant Voltage

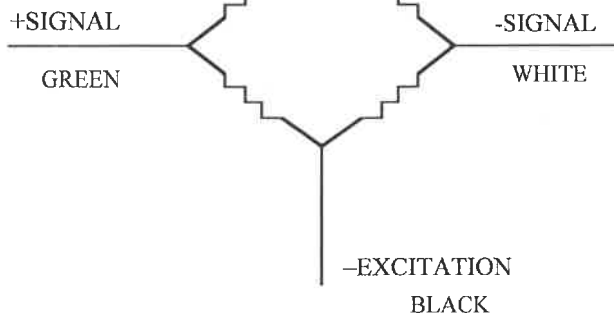
Pressure Calibration Data all readings are in mV DC						<i>Date of Pressure Calibration</i>
						19 JUN 21
Pressure	Increase	Decrease	Ideal	Linearity (%FS)	Hysteresis (%FS)	STATIC ERROR BAND ± .10% FS BFSL
0 PSIA	-.01	-.09	-.01		.08%	
1000 PSIA	52.74	52.62	52.54	.19%	.11%	
2000 PSIA	105.09		105.09			
SENSITIVITY	105.10					

Thermal Calibration Data all readings are in mV DC						<i>Date of Thermal Calibration</i>
						19 JUN 21
	Low Temp.	Ambient	High Temp	Temperature Range	Thermal Balance Shift	Thermal Sensitivity Shift
Temperature	35 °F	75 °F	75 °F			
0 PSIA	-.06	-.09	-.12	35°F to 75°F	-.03%FS	1.13%FS
2000 PSIA	103.94	105.10	105.08	75°F to 75°F	-.03%FS	.01%FS
Sensitivity	104.00	105.19	105.20	AVERAGE	± .001% FS/°F	± .014% FS/°F

Notes

DATE OF LAST MEASUREMENT:
23 JUN 21

BALANCE OUTPUT:
-.07 mV DC



INPUT RESISTANCE 1792 Ohms
OUTPUT RESISTANCE 791 Ohms

<i>Data Entered and Reviewed By</i>	<i>Date Data Entered</i>
THARIN THAV	1 July 2021